

JONQUIERE, QC, Canada

WMO: 716170

Lat: 48.4250N

Lon: 71.1425W

Elev: 445

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-15.7	-27.6	1.2	-20.2	-22.8	1.6	-15.2	27.0	14.5	25.0	9.2	6.7	280	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.1	67.5	80.8	66.1	77.6	64.9	71.1	79.2	69.2	76.6	67.3	74.5	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	106.1	75.1	66.5	99.3	72.8	64.7	92.9	71.2	35.2	79.6	33.6	76.8	32.0	74.3	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.7	18.6	DB	-27.0	90.0	5.8	3.1	-31.2	92.2	-34.6	94.0	-37.9	95.8	-42.1	98.0	(4)
(5)			WB	-27.5	74.3	5.7	2.2	-31.7	75.9	-35.0	77.2	-38.3	78.5	-42.4	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	6.2	10.0	21.7	36.4	51.2	61.0	65.8	63.7	55.6	43.5	30.4	15.9		
	DBStd	22.63	13.80	12.58	11.84	8.49	7.91	6.97	5.11	5.50	7.47	7.46	9.32	11.90		
	HDD50	5754	1358	1120	878	410	80	4	0	0	27	230	590	1057		
	HDD65	9816	1823	1540	1342	857	433	158	51	90	294	667	1039	1522		
	CDD50	1599	0	0	1	3	117	335	491	426	196	28	2	0		
	CDD65	186	0	0	0	0	6	39	77	51	13	0	0	0		
	CDH74	1597	0	0	0	2	121	420	577	371	103	3	0	0		
	CDH80	361	0	0	0	0	31	114	123	75	18	0	0	0		
	WSAvg	8.6	9.1	9.1	9.5	9.7	8.8	8.0	7.2	7.1	7.4	8.8	9.3	8.7		
	PrecAvg	37.4	2.0	1.9	2.0	2.2	3.0	3.9	4.4	3.9	4.0	3.6	2.6	3.0		
	PrecMax	42.1	3.4	3.7	4.2	3.7	6.4	6.1	7.2	8.8	8.0	5.7	5.0	4.3		
(15) Precipitation	PrecMin	32.8	0.6	0.8	1.0	1.0	1.2	1.5	1.7	1.5	2.1	1.8	0.7	1.5		
	PrecStd	3.0	0.7	0.7	0.9	0.9	1.2	1.1	1.4	1.7	1.3	1.1	0.8	0.7		
	0.4%	DB	42.1	42.6	52.3	67.5	84.2	88.7	87.7	86.4	82.4	71.1	58.7	42.1		
	MCWB	40.3	40.0	42.4	51.0	61.3	69.9	70.0	69.4	69.2	59.8	53.6	39.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	36.4	35.8	44.5	59.4	77.0	83.2	83.4	81.8	76.6	64.4	50.5	37.3		
	MCWB	35.1	33.3	38.0	45.9	57.9	65.8	68.1	68.0	66.7	56.1	46.6	35.5			
	5%	DB	31.4	32.8	40.5	53.7	71.2	78.8	80.1	78.2	72.0	58.7	45.7	34.6		
	MCWB	30.0	30.7	35.4	43.2	55.9	64.1	66.8	66.1	63.0	52.1	42.5	33.0			
	10%	DB	26.2	29.5	37.4	48.9	66.2	74.7	76.9	74.8	68.0	54.6	42.1	32.3		
	MCWB	24.7	27.6	33.8	40.9	54.0	62.0	65.6	64.2	60.9	49.6	39.4	31.0			
	0.4%	WB	40.4	40.1	44.8	53.9	66.1	72.8	74.1	72.3	71.3	61.4	53.7	39.9		
	MCDB	41.5	42.3	49.5	64.5	75.2	83.8	82.0	81.1	79.8	69.1	58.5	41.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	33.7	38.9	47.6	61.9	69.6	71.1	69.9	67.9	57.0	47.4	35.6		
	MCDB	36.4	35.2	42.7	55.5	72.0	78.1	78.3	78.0	74.1	63.0	49.8	36.9			
	5%	WB	30.1	30.8	36.3	44.6	58.0	67.0	69.4	68.0	64.6	53.3	42.7	33.3		
	MCDB	31.4	32.6	39.5	52.2	67.0	74.3	76.3	75.6	70.7	57.4	45.2	34.5			
	10%	WB	24.8	27.4	33.9	42.0	54.9	64.2	67.3	66.2	61.8	50.1	39.6	31.1		
	MCDB	26.2	29.3	36.9	47.9	64.8	71.9	74.3	72.7	67.0	53.8	41.9	32.3			
	MDBR	16.9	17.8	17.6	17.0	20.6	19.7	18.3	18.1	17.3	13.1	11.2	13.0			
	5% DB	MCDBR	19.3	16.6	19.4	26.2	29.6	27.4	23.7	23.6	23.3	21.1	16.4	13.4		
(36) Mean Daily Temperature Range	MCWBR	18.6	15.2	14.0	15.7	14.7	13.7	11.3	11.9	13.7	13.8	13.5	12.6			
	5% WB	MCDBR	19.0	16.4	16.4	22.7	24.1	21.7	19.6	19.4	20.4	18.8	15.1	13.0		
	MCWBR	18.5	15.1	12.9	15.0	14.0	12.9	11.2	11.0	13.3	13.4	13.0	12.5			
	taub	0.266	0.285	0.315	0.378	0.359	0.385	0.403	0.382	0.370	0.329	0.317	0.265			
(40) Clear-Sky Solar Irradiance	taud	2.259	2.218	2.204	2.220	2.405	2.360	2.324	2.383	2.403	2.502	2.369	2.318			
	Ebn at Noon	254	275	284	276	286	278	271	272	264	259	232	240			
	Edh at Noon	23	31	37	41	36	38	38	35	31	24	22	20			
	RadAvg	414	721	1118	1413	1636	1746	1647	1398	1071	612	332	273			
(44) All-Sky Solar Radiation	RadStd	51	95	92	110	156	117	161	93	80	63	34	30			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page